# Forcing beats fitting: the minimal generative-evidence case for BST (one page)

Why the same five integers that name one geometry force the Standard Model's dimensionless content — and how to tell forcing from fitting in one question

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## Forcing beats fitting — the one-page case

### The distinction that decides everything

A theory that *reproduces* N physical constants may still have *fitted* them. The only question that separates derivation from numerology is where the physics enters: upstream (in the premise, fixed before you look at the data) or downstream (in the output, tuned to match). Wyler (1969) reproduced $\alpha$ to high precision and is remembered as numerology — not because the number was wrong, but because his construction anchored on the target. BST's claim is that its anchor is target-innocent: the premise is one geometry, fixed by its own mathematics, and the constants fall out downstream.

### The premise (fixed target-innocently, before any observable)

One bounded symmetric domain, D_IV$^5$ = SO$_0$(5,2)/[SO(5)×SO(2)], selected by internal criteria (it is the unique automorphism-invariant Born space of the required type). Its invariants are five integers — N_c = 3, n_C = 5, C$_2$ = 6, g = 7, N_max = 137 (with rank = 2) — each a property of the geometry (root multiplicities, dimensions, Casimir), none read from a physical measurement. That is the whole input. There are zero dimensionless parameters tuned to the data, and — a theorem, not a shortfall — exactly one dimensionful input (the electron mass; a dimensionless set cannot produce a quantity with units, so one scale is the floor every theory pays, GR with G, the SM with v). We name the one open question at the premise itself: the domain's own dimension, n_C = 5. The selection criteria narrow the domain sharply, but whether n_C = 5 is fully *forced* or enters as a primitive label — the honest referee's "where is this five in the definition?" — is the frontier of the premise, and we mark it rather than bury it. (The five *derived* integers are properties of the domain once chosen; this question is one level up, about the choice.)

## The minimal generative evidence (a few sharp, target-innocent, over-determined)

| observable | forced form (geometry only) | value | status |
|---|---|---|---|
| m_p/m_e | $6\pi^5$ = N_c!·$\pi$^{n_C} | 1836.12 vs 1836.15 (0.002%) | Derived |
| $\alpha^{-1}$ | N_max = N_c$^3$·n_C + rank | 137 (+0.036 curvature) | Identified-strong |
| Newton's G | reduces to m_e + $\alpha$ (2 inputs → 1) | 0.065% | Derived-structure (one relation, several readings — counted once) |

The G reduction rides on $\alpha$ through 24 powers of it (m_Planck/m_e = $6\pi^5\cdot\alpha^{12}$; G $\propto$ m_Planck$^{-2}$). That $\alpha$ is the *Identified* $\alpha$ of the row above — measured atomically (recoil, g−2), G-independent — not derived: the reduction is real and G-independent, but the coupling it rides on is a measured number, and we tier it as such.

These are not three lucky hits. The same five integers force these and ~150 more, across independent sectors (nuclear, electroweak, PMNS/CKM, cosmology). When one integer fixes many observables, that is a single constraint with many readings — a web, not N independent fits — and it is the opposite of what a fitted model can do (a fit spends one parameter per number). We state this up front, per our own discipline, so a "new" prediction is checked as *another reading of the same object*, not a fresh win — and, just as strictly, we flag where the web is *not* yet one object. (Example, at its honest limit: for the charged leptons the three mass *numbers* satisfy one exact relation — Koide's Q = 2/3, to 0.001% — but that relation is currently *identified*, not yet derived, and the individual mass *forms* are separate identifications, not readings of a single equation. The muon's $(24/\pi^2)^6$ is the one open form; deriving it and the Koide filling-amplitude would close the charged-lepton sector on the electron mass alone. We do not double-count the tau's 49·71 form — a mass the mass-chain already yields. A patchwork we call a patchwork.)

## The one-question test a reviewer can apply

For each claimed derivation: does its *derivation* reference the target value, or only the geometry? If only the geometry (target-innocent integers, no look-elsewhere), it is forced. If the target appears in the anchoring, it is fitted, and we label it so. Every load-bearing claim in BST carries this provenance tag (Derived / Identified / Conditional / Structural).

## The instrument is trustworthy because it cut its own count

The discipline that produces these labels is falsifiable *in use*: applying it lowered our own honest "derived" count (21 → 12) while the accuracy of the numbers never moved. A device that reduces its makers' claims and keeps the data fixed is a working instrument, not an advocacy tool — most of the rules were written from a case where the rubric caught its own authors mid-over-claim. We publish our open ledger (every soft spot at tier) and our honest boundaries — and behind the sharpest one is now a theorem. Decomposing the finite internal space of the geometry under its symmetry K = SO(5)×SO(2) is exact: it splits into three pieces landing on the three real division algebras (Frobenius: $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ are the only ones) — a real three-dimensional color space, a complex charge circle, and a quaternionic weak spinor. Count the internal gauge generators each supplies: {0, 1, 3}, and 0 + 1 + 3 = 4 = dim(U(1)×SU(2)). The electroweak group's dimension is a *count*, not a fit — a one-line check a referee can do in a margin. U(1) and SU(2) are DERIVED (the complex and quaternionic blocks carry an intrinsic complex structure, so their unitary symmetries are geometry-given). The color block contributes zero generators: there is no internal color group. Its real 3-space gives only a *geometric* SO(3) rotation of the three color directions (spacetime-like, and the reason N_c = 3), not an internal gauge factor — so SU(3) is not part of the geometry's internal algebra at all; it is the standard gauge principle, the same input every derivation of the Standard Model takes. That asymmetry is also *why* BST forbids grand unification: the three factors are not on equal footing (two internal, one none), so there is no reason to unify them. Two honest notes: the same three-color object carries the integer N_c = 3 that appears in the weak mixing angle ($\sin^2\theta$_W ≈ 3/13, ~0.2%; a loose tie of a shared object, not a forced angle); and chirality — the world being left-handed — comes from hypercharge, not the weak block (an SU(2) doublet alone is vector-like; Y ≠ 0 is what makes it chiral). The credential is theorem-backed: BST does not *assume* the Connes finite algebra $\mathbb{C}\oplus\mathbb{H}\oplus M_3$ — it reads it off one geometry, and tells you why the electroweak group is exactly four-dimensional and why color has no internal group. We claim exactly that — electroweak derived, color the gauge-principle input, with the reason — and never "the geometry forces the dynamics."

## The exposure (generative theories forbid; fits don't)

BST makes forbidding predictions — the Five-Absence set: no supersymmetry, no proton decay, no dark-matter particle, no magnetic monopoles, no sterile neutrinos, no grand unification. Each follows from the irreducibility of the one geometry; any single positive detection falsifies the framework. A fitted model can always absorb a new particle; a forced one cannot. That asymmetry is the point.

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*Companion to Paper A (the derivations) and Paper B (the full 17-rule Forcing+Evidence standard). This one page is the minimal case: premise fixed target-innocently, a handful of sharp forced numbers, the over-determination web,* *the one-question test, the count-cutting instrument, and the forbidding falsifiers. Reproduction is not derivation; the difference is upstream vs downstream, and it is checkable in one question.*